

Ansh Madaan

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Education

Thapar Institute of Engineering and Technology 2023 – Present
B.E. Computer Science CGPA: 9.16

Govt. Boys Senior Secondary School 2022 – 2023
Class XII 95.4%

Experience

Summer Intern, TIET ELC (Experiential Learning Centre) Jun 2025 – Jul 2025

- Worked on EEG signal analysis for schizophrenia detection using deep learning, as part of a summer internship at TIET's Experiential Learning Centre.
- Preprocessed raw multi-channel EEG recordings and benchmarked classical ML baselines to establish a performance floor ahead of deep learning experiments.

Projects

Career Ops — AI-Powered Job Application Agent Live Demo | GitHub
Python, FastAPI, Next.js, Supabase (PostgreSQL), Google Gemini API, TOPSIS, asyncio May 2026 – Present

- Built a **FastAPI** platform that aggregates **6,000+ listings** across four ATS providers (Greenhouse, Lever, Ashby, Internshala) and ranks them down to the **top ~50 roles** using **TOPSIS multi-criteria scoring**.
- Built an **async scraping engine** (Python `asyncio`) for concurrent, real-time multi-source scraping; extending with **AI resume tailoring**, **ATS-optimized PDF export**, and **one-click auto-apply**.
- Built a **Gemini + Pydantic parsing pipeline** that turns raw resumes into validated candidate profiles for automated **skill extraction** and job matching.

EEG-based Schizophrenia Detection using Deep Learning GitHub
Python, TensorFlow/Keras, scikit-learn, XGBoost, Grad-CAM, Integrated Gradients Jun 2025 – Jul 2025

- Designed a hybrid **CNN–LSTM with attention** reaching **97% accuracy** on a clinical EEG dataset (**84 subjects**) under a **subject-wise split**, outperforming CNN-only and LSTM-only baselines by **6–9%**.
- Preprocessed multi-channel EEG via **bandpass filtering (1–50 Hz)** and temporal windowing, improving signal quality for downstream classification.
- Interpreted predictions with **Grad-CAM** and **Integrated Gradients** to surface the channels and time-segments driving classification; **paper accepted at ICDSP 2025**.

Voice Assisted Terminal GitHub
Python, Google Speech Recognition, Google Gemini API Jan 2024 – Mar 2024

- Built a voice-controlled terminal that converts spoken natural language into shell commands using **Google Speech Recognition** and the **Gemini API**, enabling hands-free command execution.
- Packaged and published the tool on **PyPI** with a console entry point for one-command `pip install` and cross-platform use.

Technical Skills

- Languages:** C/C++, Python, SQL, R, JavaScript, HTML, CSS
- Frameworks & Libraries:** TensorFlow, scikit-learn, FastAPI, Pydantic, OpenCV, NumPy, Pandas, Matplotlib
- Databases:** MySQL, MongoDB
- Developer Tools:** Git, Postman, Jupyter Notebook, PyPI, Figma, VS Code
- Core Concepts:** DSA, OOP, OS, DBMS, Computer Networks, Software Engineering, Deep Learning

Achievements

- Solved **600+ LeetCode problems** across **dynamic programming, graphs, trees, and arrays**.
- General Secretary**, ACM Student Chapter, Thapar Institute — leading a team of **200+** members.
- Winner, **ACM Projectathon** — led a team to victory among 110+ teams.
- Recipient of **Merit Scholarship** for academic excellence (Top 10% at Thapar Institute).